

Diwas Kharel

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PROFILE

Early-career AI engineer with hands-on experience building LLM assistants, evaluating model outputs, and developing computer vision systems. Seeking to bring Python development, evidence-based retrieval, and benchmarking experience to the Researcher role at the hiring organization.

SKILLS

Python · LLMs · LangGraph · LangChain · RAG · Hybrid retrieval · Embeddings · FAISS ·
Computer vision · Image segmentation · Model evaluation · Benchmarking

EXPERIENCE

AI Research & Industrial AI Assistant Intern

Apr 2026 – Jun 2026

Vaski Group Oy

- Built an industrial RAG assistant that answers operator and maintenance questions using official manuals and training-video transcripts.
- Developed PDF and video ingestion with section-aware chunking and source metadata, then combined semantic search, BM25, and reranking before GPT-4.1 synthesis.
- Enforced source citations and evidence-based refusals for unsupported answers about safety circuits and machine-reset procedures.
- Specified and validated CPU-only, on-premise deployment and designed an air-gapped variant using a local Ollama model.
- Authored an internal technical report and deployment guide covering architecture, hardware, installation, and local-versus-cloud trade-offs.

LLM & NLP Recommendation Systems Intern

May 2025 – Jul 2025

RightProb Solutions

- Built a bidirectional semantic matching engine for candidate profiles and job vacancies, spanning data preparation, embeddings, indexing, retrieval, and interface.
- Selected multilingual-e5-base for Finnish, English, and mixed-language records, and implemented cosine-similarity retrieval with FAISS.
- Indexed 2,184 profiles and 1,000 job postings and delivered ranked matches in both directions.
- Built a recruiter-facing CLI displaying scored top-k results to evaluate ranking quality and iterate on embeddings.

AI Tools & Text-to-Speech Intern

Nov 2025 – Feb 2026

Vaski Group Oy

- Benchmarked commercial and open-source AI voice engines for quality, cost, language coverage, and licensing.
- Presented a recommendation based on benchmark findings that the documentation team adopted.
- Validated voice-over output against editorial standards and established a repeatable production workflow for technical training material.

PROJECTS

Agentic Data Analyst [GitHub](#)

Python · LangGraph · LangChain · E2B · Pydantic · Streamlit

- Designed a LangGraph pipeline with conditional routing for exploratory analysis, supervised prediction, and time-series forecasting.
- Built LangChain ReAct agents that execute and iterate on analyses in a shared, persistent E2B Python sandbox.
- Implemented dataset validation, feature preparation, model evaluation, and forecasting phases with typed Pydantic state and result schemas.
- Created a separate LLM verification pass that checks report claims against ground-truth analysis state.
- Built a benchmark harness using three synthetic datasets to test routing, state validity, and model performance against baselines.

Image and Shape Detectable Pick and Place [GitHub](#)

Streamlit

- Built a camera-guided pick-and-place system that detects objects and maps their image coordinates to a Dobot MG400 robot workspace.
- Segmented objects using thresholding and connected components, extracting masks, bounding boxes, and centers.
- Detected circles, squares, and rectangles from contours and classified object colors using masked HSV values.
- Implemented planning and automated execution modes, with a Streamlit interface for reviewing detections and coordinates.

EDUCATION

Bachelor of Engineering ICT Robotic and AI

Häme University of Applied Sciences, HAMK